Power7 — Online Store Setup & Planning

Energy Drinks, Pre-Workouts, Coffee & Natural Vitamin Juices

Presentation 1 · E-Commerce Semester Project

1. Team & Roles

The project is developed by a team of two students, covering full-stack responsibilities across all project modules (A1–A12).

Team Member	Role / Focus Area	Assigned Requirements
Ponoran Claudiu Mihai	Storefront & User Management Catalog, Search, User Accounts, Order History, Responsive Design, Automated Tests	A1, A2, A4, A8, A11, A12
Oprea Iulian Marian	Purchase Flow, Payment & Administration Shopping Cart, Checkout, Webhook Payment, Stock, Admin Panel, Security	A3, A5, A6, A7, A9, A10

2. Context: Niche, Target Customer, Competitor Analysis

Product Niche

Power7 specializes in premium energy-boosting products, organized into 4 distinct categories:

- **Energy Drinks:** Sugar-free and classic energy boosters, functional focus drinks.
- **Pre-Workout Supplements:** High-intensity endurance formulas, pump formulas, and amino complexes.
- **Specialty Coffee:** Single-origin whole beans, cold brews, and high-caffeine artisanal roasts.
- **Natural Vitamin Juices:** Cold-pressed fresh fruit juices, immunity shots, and organic vitamin blends.

Target Customer

Young adults, students, athletes, gamers, and working professionals aged 18–35 who demand rapid energy solutions, clear ingredient transparency, mobile-first ordering, and fast checkout.

Competitor Analysis (3 Real Stores)

Store Analyzed	What We Adopt	What We Do Differently
GymBeam	Detailed nutritional value tables, clear caffeine/dose indicators, category filtering.	Simpler, focused 4-category catalog without clutter; specialized express checkout flow.
MyProtein	Step-by-step cart overview, clear pricing breakdown, discount/promo presentation.	Faster and more optimized UI, with 0 forced subscriptions.
5togo / Coffee Stores	Focus on fresh roast descriptors, origin transparency, clean visual presentation.	Full integration of energy drinks + pre-workout + coffee under one unified modern store.
G Fuel	Clear profiles of flavours and multiple bundle offers for customers.	A simpler and cleaner design of the website.

3. Technology Stack & Justification

Frontend: HTML5 + Bootstrap 5 + JavaScript (ES6)

- Utility-first and responsive grid layout built with Bootstrap 5, combined with vanilla JavaScript for client-side interactivity (cart updates, search filtering, dynamic UI).

Backend: ASP.NET Core 10 (.NET 10 / C#)

- Enterprise-grade web framework offering built-in security, dependency injection, routing, Model Validation, and CSRF Anti-Forgery Token protection.

Database: SQLite/PostgreSQL + Entity Framework (EF) Core

- Migration tracking, parameterized SQL queries, and optimistic/pessimistic concurrency handling.

Payments: Stripe Checkout (.NET SDK)

- Server-side webhook handling via ASP.NET Controllers to process payment state transitions asynchronously.

Testing & Environment: xUnit + Docker Compose

- Containerized deployment using official **.NET SDK** and Database images, enabling single-command execution (**docker compose exec app dotnet test**).

4. Mandatory Requirements Breakdown (A1–A12)

Code	Requirement Description	Owner	Target Week
A1	Product Catalog (Paginated, 4 categories, detailed product pages)	Ponoran Claudiu	Week 4
A2	Basic Search (Text search on product names with empty state handling)	Ponoran Claudiu	Week 5
A3	Shopping Cart (Add/update/remove items, persistent for logged users)	Oprea Iulian	Week 5-6
A4	User Accounts (Registration, login/logout, password hashing, saved addresses)	Ponoran Claudiu	Week 4
A5	Checkout Flow (Address selection,	Oprea Iulian	Week 7

	delivery methods, order summary)		
A6	Sandbox Payment (Stripe webhook confirmation, async status updates)	Oprea Iulian	Week 8
A7	Stock Management (Decrease on payment, restore on cancellation, lock check)	Oprea Iulian	Week 9
A8	Order History (Customer order viewing with full details and status)	Ponoran Claudiu	Week 8
A9	Administration Panel (Product/category CRUD, order management & status transitions)	Oprea Iulian	Week 5, 8, 9
A10	Basic Security (Server-side validation, CSRF, server-enforced RBAC)	Oprea Iulian	Week 6, 7, 9
A11	Responsive Design (Mobile-first catalog & checkout tested on 360px)	Ponoran Claudiu	Week 6, 7, 9
A12	Automated Tests (≥10 xUnit tests covering cart, total calculation, payment)	Ponoran Claudiu	Week 10

5. Weekly Implementation Schedule

Week & Period	Objective & Deliverables	Owners
Week 3 (Oct 12–18)	Setup: Git repository, Docker Compose, HTML + CSS + JavaScript skeleton, README	Both
Week 4 (Oct 19–25)	A1 Catalog, A4 User Accounts & Address Schema, Initial .NET setup	Ponoran Claudiu
Week 5 (Oct 26–Nov 1)	A2 Search, A9 Admin CRUD (products/categories), A3 Cart (add items)	Both
Week 6 (Nov 2–8)	A3 Cart completed, A10 Validation & CSRF, A11 Responsive Catalog	Both
Week 7 (Nov 9–15)	A5 Checkout flow, A10 Admin access security, A11 Responsive Checkout	Oprea Iulian
Week 8 (Nov 16–22)	A6 Stripe Payment Webhook, A8 Order History, A9 Admin Order List	Both
Week 9 (Nov 23–29)	A7 Stock Management & concurrency lock, A9 Order status flow, A11 360px check	Oprea Iulian
Week 10 (Nov 30–Dec 6)	A12 Automated tests (≥10), stock concurrency test, demo seed data, feature freeze	Ponoran Claudiu
Week 11–12 (Dec 7–20)	Integration testing, final README, server deployment	Both

	demo, presentation rehearsals (P3)

6. Top 3 Risks & Mitigation Strategies

• **Risk 1: Stock Race Conditions (Last Item Over-purchasing):** Mitigation: Use database row locking (pessimistic locking) inside a transaction and write a dedicated concurrency unit/functional test.

• **Risk 2: Merge Conflicts & Codebase Divergence:** Mitigation: Enforce feature-branch workflow with single-owner pull requests and code review requirements before merging to main.

• **Risk 3: Environment Discrepancies ('Works on my machine'):** Mitigation: Standardize deployment and execution environment strictly using Docker Compose with fixed **.NET 10 SDK and PostgreSQL/SQLite** images.

7. Repository Structure & Way of Working

GitHub Repository: https://github.com/DarkWarDark/Power7

Initial README

Project description, team roles, local setup instructions via Docker Compose, and test execution commands

Teacher access

Teacher has been added to the repository.

Git flow

One branch per requirement, a PR with one reviewer is mandatory, main stays stable.

Tracking

One issue for each requirement A1–A12, with owner and week.